E-304 — guided full-length row/column clutch

PRINT TEST: DRAFT — unresolved pin-wall toolpath gaps; physical operation untested.

Owner-directed parallel mechanical investigation, 25 October 2026, based on commit 1071f695b93951246a2865038e7fb9de9bdf2a05. **DES-009 remains the active primary design and is unchanged.** No physical E-304 test has been performed. Completion means the investigation and one DRAFT physical-validation package, not a qualified mechanism. The configured 3MF reopens/reslices successfully, but unresolved approximately 0.207-mm pin-wall toolpath gaps prevent print readiness.

Decision

Select **one guided 30° friction cone per pixel**, with a captured full-length row cam, a full-length column ceiling gate and one integral compliant follower. A permanently meshed common rack drive turns the inputs. Two stationary journals locate each input and each output; the moving selectors supply force and logic, not shaft alignment. This is the strongest practical candidate for one test, not authorization to build 80×80.

Owner question	Result and evidence level
What improves on E-297–E-300?	Sliding double-D torque fit removed; input gear/shaft/cone integrated; output cup/screw integrated; two stationary guide planes per shaft; lower thrust-contact radius; local compliance for unequal contact heights; deep guided selector beams and positive withdrawal. These are design/CAD improvements, not measured reliability gains.
Components per pin?	Four printed moving bodies including the terrain pin: pin, output screw/cup, input gear/cone, follower. Zero purchased parts per pin; one integral spring function. Four radial bearing lands, two pin guides and axial/withdrawal contacts are still repeated functions in shared structures.
Arbitrary initial angle?	An ideal coaxial cone makes contact at every real relative angle without deliberate indexing .1,440-angle representative tooth-envelope analysis and 72-angle actual cone mesh sweeps are supplied. Printed friction, seam, eccentricity, tilt and drag remain unproved. This is not a universal rejection of toothed couplings.
Full-length stiffness?	Longitudinal 2.54-mm selector motion permits fixed guides every 25.4 mm. Use a closed printed box/cam beam with continuous inserted stock; two 2-mm steel rods are the conservative default at the calculated force; carbon saves weight but needs measured modulus and joint evidence. Loose rods do not earn bonded-composite stiffness. Steel and other stock are compared. No separately powered short rows.
Plausible at 5.08-mm pitch?	The actual 4.4-mm pins and retained 3.2-mm/3-mm-lead thread fit in a complete neighboring 2×2 CAD model. Cup gap .48 mm and gear gap .88 mm. The fixture is 80.2 mm overall at neutral (pin top Z79 mm) and tests±1.5-mm travel. Full-array assembly, long-stroke drive, guide manufacture and economic advantage remain unresolved.
Torque/force?	Nominal linear follower model predicts 1.10 N selected, 1.79 N half-selected. A stated friction/drag case predicts .400 Nmm net versus E-303's assumed .343 Nmm moderate start; less favorable friction removes that margin. A fully loaded 80-cell row can require roughly 113–183 N at the chosen cam friction. These are calculations, not measured loads.
What remains unknown?	Actual clamp force, cone grip and release, radial rubbing (adverse guide/axial-clearance corners can interfere), creep/fatigue, unselected height retention, small printed gear quality, assembly yield, neighbor isolation under obstruction and large-array maintenance.
Single continuation result?	All four cells repeatedly engage from sampled stationary phases, move bidirectionally under the stated light/moderate load and release at one setting; half-selected cells stay still and one restrained output does not stop the others. Seek that result once, then await an explicit owner decision.

Evidence and delivery

Downloadable engineering report PDF combines the investigation, drawings, verification limits and owner test instructions.

- Previous failures and architecture/engagement comparison: physical owner observations separated from CAD and untested hypotheses; single cone, flat face, guided teeth, two clutches in series, collet and wrap alternatives.

- Full-length selector study: bending, torsion, local shear, buckling, axial shortening, supports, manufacturing, rod joints and thermal effects.

- Concrete mechanism, tolerances, force paths, component and cost budget. Broad full-system screening allowance€1,300–3,800 excludes labour and is not a vendor-qualified actuator/BOM quote.

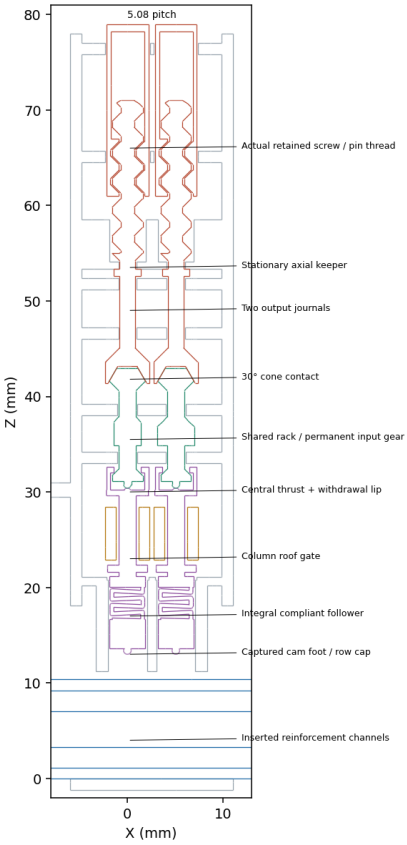
- Primary sources and observed stock prices, reproducible calculations/CAD/checks, verification record.

- One owner test package, including actual STLs, configured X1 Carbon Bambu 3MF, tolerance samples, shared drive, drawings and blank results form. Its explicit status governs printing; a generated archive alone is not READY.

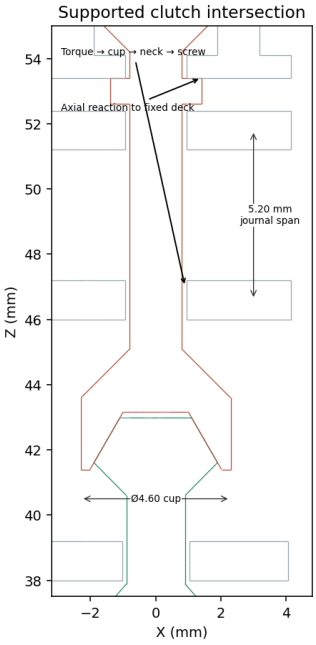
Stop: one meaningful validation package prepared. No issue, purchase, automatic print campaign or further development is initiated. Owner feedback and an explicit decision are required before continuation.

Dimensioned sections from delivered CAD

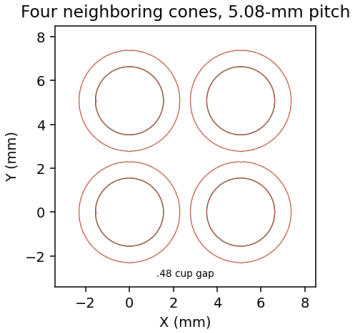
Loaded 2×2: section through first row
Actual nominal meshes; elastic follower is a linear model



Supported clutch intersection

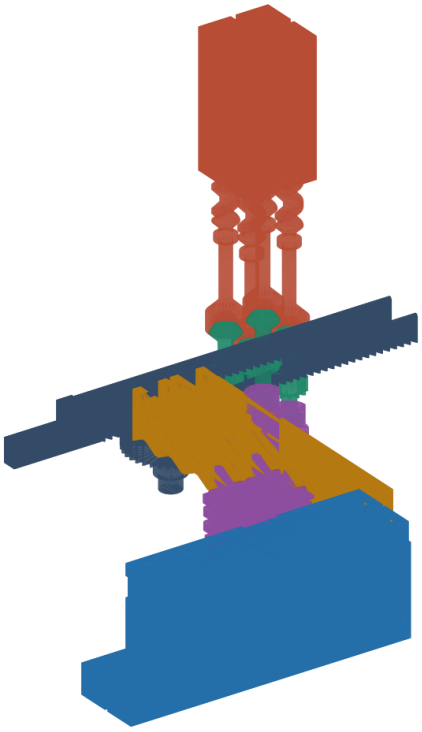


Four neighboring cones, 5.08-mm pitch

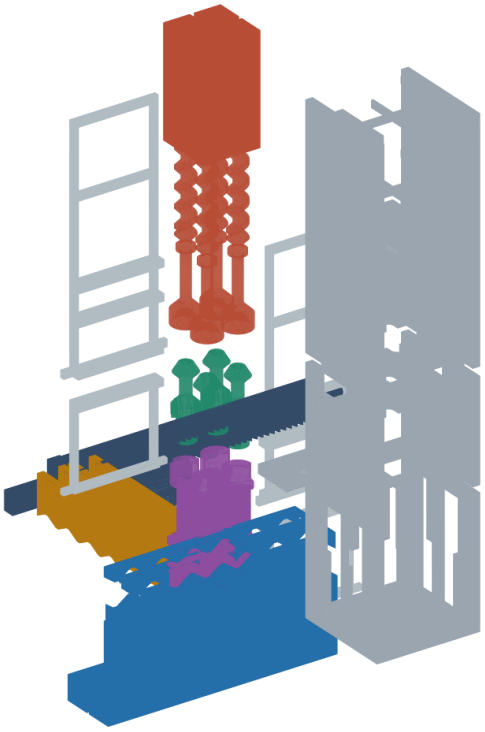


Operating assembly and exploded layout

2×2 mechanism, stationary frame hidden for visibility



Exploded layout: shared covers, open cam caps, resident shafts



Blue: row bases/caps • gold: columns • purple: guided elastic followers • green: input rotors
Red: supported screws and terrain pins • grey: stationary frame/covers • navy: one common rack drive

E-304 — evidence, alternatives and selection

Owner-directed parallel research, 25 October 2026. Repository baseline 1071f695b93951246a2865038e7fb9de9bdf2a05. **DES-009 remains active and is unchanged.** No terrain, scheduling or update-scenario research is repeated. No independently powered short rows are proposed.

What actually failed

The decisive physical record is the complete E-298 owner report, including its later stop-block update. Its observations outrank the earlier experiment summaries saying “not yet tested.”

Evidence	What it establishes	What it does not establish
E-295: five compact pairs, three nominal 0.175-mm relief pairs reportedly very smooth	This specific 3.2-mm, 3-mm-lead, 90° thread in a 4.4-mm pin can print and move by hand	Measured running/start torque, self-locking, array yield or long stroke
E-298 owner: partial kit, auto-orientation and auto-placement used; four unspecified parts not printed	Assembly was incomplete; saved orientation cannot reconstruct the print	A controlled comparison of the supplied nominal variants
Owner: slender screws produced spaghetti; brim cleanup awkward	The actual unsupported screw print and cleanup failed	Failure of the thread principle or all upright screw prints
Owner: points directly opposite blocked; only one pair tested	This printed contact pair did not self-align at that alignment	Universal failure of every pointed, toothed, asymmetric or compliant coupling
Owner: selectors already bent; limited travel prevented complete closure; notches bent easily	Global bending and local notch deformation independently reduced engagement	Exact cause of initial warp, modulus, force or dimensional error
Owner: substantial wobble; shallow hand socket wore quickly	The assembled constraint chain and test input were inadequate	Which individual bearing clearance caused what fraction of wobble
Later owner update: stop blocks used, one pin moved up/down smoothly	Positive one-pin motion after manual assembly/setting	Automatic engagement, simultaneous selection or neighbor isolation
E-299	Wider dogs, 3-mm stroke, larger holders, screw braces, split frame exist in CAD/slicer	No completed E-299 physical report was found
E-300	45° cones, stronger vertical section, two-contact rocking row; calculated thrust loss	No physical cone result was found; a blank form is unknown, not a failure
E-301 / DES-009	One compact pin moved using a microstepper and improvised tape coupling	A traveling bank, automatic docking or this new mechanism

Actual geometry inspected: E-297 cad/generate.py, E-298 contact-analysis.md and point geometry, E-299 CAD/report, E-300 cad/generate.py, parameters.json, actual mesh-derived section drawing, and the retained E-295 thread/pin STLs. The old double-D is Ø2.2/AF1.6 in an approximately Ø2.5/AF1.9 socket. Its axial sliding fit, rotor capture play and incomplete guide sectors accumulate tilt; selector flanges cannot substitute for two spaced stationary journals. E-297's 0.96-mm comb blades and E-300's 0.46-mm spine expose the local weakness at the openings. E-300's taller section improves one bending axis while weakening another; height alone does not establish torsional stiffness.

The new design removes the sliding torque spline entirely. The input gear, shaft and male cone are one part. The output cup, neck, thrust collar and screw are one part. Round stationary guides locate each shaft; selector movement never supplies radial location. This addresses a specific tolerance chain, not an assertion that double-D couplings are generally unusable.

Candidate architectures

Counts below include the resident pin/screw and the local input distribution rotor. Shared selectors, guide decks and bus are additional. A printed flexure remains a moving mechanical function even when no separate spring is bought.

Candidate	Concrete torque/support arrangement	Printed parts/cell	Purchased parts/cell	Local functions and main tradeoff
A1: guided single cone, cam + roof gate, integral compliant follower **selected**	One axially moving input rotor in two stationary journals; fixed output cup/screw in two journals and captured thrust collar; follower has central thrust pad and withdrawal lip	**4**	**0**	Input spin + axial shift, output spin, pin translation, follower cam motion, elastic differential travel. One elastic function. Conical traction and guide drag unmeasured.
A2: same guided selector, flat annular face	Same four parts and guides; no wedge effect	4	0	Phase-independent geometry and simpler release, but half the gross torque of the 30° cone at equal wear radius/force. Best fallback if cone sticking dominates. No extra prototype prepared.
A3: guided sliding dog/face clutch	Fixed driven shaft, sliding keyed sleeve, output screw, compliant gate follower, pin	5	0	Positive torque, less sustained clamp force after seating; one sliding torque fit returns. Tested representative teeth retain blocked phases or resistance-dependent self-indexing.
B: two independent clutches in series	Fixed input rotor □ row-controlled keyed sleeve □ independently supported intermediate shaft with two clutch faces □ column-controlled sleeve keyed to output screw; two guided followers; pin	**8**	**0**	Two elastic functions; intermediate shaft needs two radial journals and two thrust faces. Each selector can engage a simple independent clutch. Phase-independent if both are frictional. No logical roof gate; more assembly and two slip interfaces.
C1: bidirectional radial collet	Shared driven cage, radially closing slotted collet on screw tail, tapered axial sleeve and row/column gate	At least 5	0–1	Continuous angular grip and possible torque multiplication; thin slotted fingers, radial pressure and release hysteresis become repeated functions. At 5.08 mm, guide/cage/collet walls compete for the same radius. Not disproved; less convincing FDM first test.
C2: bidirectional wrap/roller wedge clutch	Opposed wraps or opposed roller paths, supported race, controlled release	At least 5 plus wraps/rollers	Several	Compact torque possible, but direction reversal and release add repeated precision contacts. A one-way bearing is not a bidirectional selector.

Candidate	Concrete torque/support arrangement	Printed parts/cell	Purchased parts/cell	Local functions and main tradeoff
C3: row-level upstream clutch plus column clutches	One row clutch feeds a full-length rotating bus, each column enables its local output	~4 plus row hardware	Bus-dependent	Can amortize the first clutch; continuous row transmission and its off-drag still require a concrete compact right-angle drive. Retained as a comparison, not silently substituted for the requested full-length mechanical selectors.

B is a genuine alternative: it replaces a mechanically gated follower with two independent ordinary engagements, and can tolerate a poorly functioning gate better than A. Its disadvantage is quantified repeated hardware and support, not merely "two is worse than one." For a series path, torque capacity is the **smaller** capacity; the two interfaces do not add torque. Both normal-force systems, all journal friction, and the intermediate shaft count. Approximately 8–12 mm additional active stack height is required even before increasing journal separation. The selected A fixture is already taller than E-300; more height is possible but not free.

Engagement investigation

Executable model samples **1,440 relative angles over 360°** for four explicitly defined three-tooth height profiles: flat dogs, triangular crowns, rounded sinusoidal crowns and asymmetric ramps. It computes

$D(\varphi) = \max_\theta [h(\theta) + h(\varphi-\theta)]$.

These are reproducible representative profiles, not claimed replicas of every historical solid. E-298's separate mesh-envelope calculation remains the evidence for its actual two-point parts. New solid cone geometry is checked separately by the mesh review.

A fixed angular phase cannot move axially past an overlapping solid volume. Allowing either rotor to turn under contact changes that condition: on a descending separation envelope axial force can do rotational work. For a single ideal contact at radius r, slope $s=|dD/d\varphi|/r$, a Coulomb screening estimate is $T/F = r(s-\mu)/(1+\mu s)$, clipped at zero. Real multiple contacts, shaft resistance and FDM steps can only be assessed with a fuller contact model or a test; this equation is not an automatic-engagement proof.

The representative envelopes have maxima or plateaus. At a smooth extremum, $dD/d\varphi=0$, so axial force supplies no first-order indexing torque. At a pointed maximum, opposing admissible contact normals can support axial force without selecting a rotational direction. A wider printed tip creates a finite plateau. Asymmetric crowns move or change these regions; the model does not assume that they eliminate them. Friction and a resisting screw widen the problematic set. This result applies to these rigid, coaxial, axial-insertion profiles and their permitted motion. **It is not a theorem excluding all toothed, guided, multi-stage or compliant mechanisms.** A tooth design with an independently imposed indexing path has changed the kinematics and deserves its own analysis; it cannot be credited with zero indexing motion.

The selected cone is a surface of revolution. In ideal coaxial geometry its first-contact lift is **0.60 mm for every real relative angle**, not just the sampled angles. No angular search, purposeful input rotation or output indexing is required to make geometric contact. Static closing therefore need not change screw height. This establishes **phase-independent contact geometry**, not torque capacity or reliable physical engagement. FDM seam, ovality, concentricity, tilt, contamination, contact pressure and friction remain phase-dependent physical effects. A shared stationary input is used during the angular engagement test; hand-setting each cup is explicitly disallowed.

At a 30° half angle, the simple taper self-wedging screen is $\mu \geq \tan(30°)=0.577$. The assumed 0.10–0.40 range lies below that boundary, but printed ridges, wedged debris, adhesive wear and tilted multi-point contact need not obey a smooth-cone Coulomb model. Both row return and column roof motion provide a positive withdrawal path through the captured follower/rotor flange. They are tested after both directions of torque; gravity alone is not counted as release.

Reversal changes friction direction without requiring another tooth phase. Slip can occur in either direction. Slip at one obstructed output allows the common input to continue only if the motor/bus has enough torque and the clutch

releases below damaging load. An obstruction in the **cam track itself** can still jam a whole row; local compliance is not universal fault isolation.

Selection criterion

Hard requirements were: retain full-length row/column control; stationary radial/tilt support; continuous-angle contact without intentional phase searching; bidirectional operation; explicit mechanical AND; no uncounted per-cell purchased bearings/springs; and a useful FDM-scale test. A1 meets these as a concrete, testable design. A2 meets them but requires approximately twice the gross pressure. B meets the phase and logic requirements with friction faces but doubles local printed part count. A3 trades clamp pressure for an unresolved engagement condition. C1/C2 add more difficult repeated interfaces before offering a clearer test.

Select A1 for one validation, not for an 80×80 build. If moderate torque cannot be transmitted in a pressure band that also gives clean release and neighbor isolation, do not continue merely because its CAD looks phase-free. The owner decides whether to continue after the single focused physical result.

Full-length selectors: structure and manufacture

The design uses one **continuous control member per complete row and per complete column**. Printed sections may join on continuous reinforcement, and stationary shoes may support them. None of these joints or supports has its own motor, address or selection state.

First resolve the boundary conditions

A vertically lifted beam cannot receive fixed intermediate support in that same vertical direction without an additional following mechanism. Quoting a short supported span for such a beam would be false. E-304 instead slides each selector **along its length by 2.54 mm**. Repeated cam tracks convert this movement to vertical follower motion. Stationary shoes now carry transverse contact load while permitting longitudinal travel. This makes short support spacing mechanically meaningful.

The field reference is 80×80 at 5.08 mm: **406.4 mm overall pitch allocation**, 401.32 mm between extreme pin centers. The structural model uses 406.4 mm, conservatively. The new stack is taller because force distribution and journals have their own space. The bench CAD reaches about 79 mm at the neutral pin top versus about 57 mm in E-300's two-pin assembly. This is roughly 22 mm additional bench height, not an established complete-display depth. No project height limit rules it out; handling, access and an eventual 40-mm pin travel still need a complete machine layout.

Comparison mechanics

$I=\int z^2 dA$, $\delta_{point}=PL^3/(48EI)$, and $\delta_{uniform}=5qL^4/(384EI)$ are used for simply supported spans. The point-load case places one obstruction at midspan. The uniform case uses 0.3–1.8 N/cell as a plausible selection-force range, **not measured load**; the comparison CSV normalizes to 1 N/cell and a separate 3 N obstruction. Adding both is conservative if the obstructed cell already belongs to the distributed load. End-fixed joints would be stiffer but are not assumed.

Closed thin-wall torsion uses $J\approx 4Am^2/\Sigma(l/t)$; open sections use $J\approx \Sigma bt^3/3$. Actual tiny sections are not very thin, and FDM shear modulus is uncertain, so these are comparative screening values. Axial shortening for a uniformly loaded selector pulled from one end is $\Delta\approx F_{total}\,L/(2EA)$; a single end obstruction uses FL/EA. Buckling uses $P_{cr}=\pi^2 EI_{weak}/L_{support}^2$, with no claimed postbuckling capacity. For a tension-driven ON stroke buckling is irrelevant to that stroke, but can matter when pushing the return or a jammed bar.

All section results, load/force sweep, support-spacing plot. PLA screening values: E=2 GPa, G=0.74 GPa, density 1.24 g/cm³; modulus sensitivities and creep are not material certificates.

Structure	Bending / torsion / manufacturing result
Flat 4.6×1.6-mm strip	At 25.4-mm support spacing, about 0.340 mm distributed deflection plus 0.326 mm for the 3 N point case. Too flexible for the selected approximately 0.15-mm deformation allowance even on short spans.
Solid 4.6×10-mm rectangle	About 0.0014 +0.0013 mm at 25.4 mm;23.2 g per 406.4-mm member before cams. Stiff, simple, but wasteful and prone to long-print warp.
Closed 4.6×10×0.8-mm box	About 0.0023 +0.0022 mm;10.5 g before cams. Better material use and torsion than an open rib or truss. Can carry guide flats without opening the main load-bearing section at every pin.
I section	Similar strong-axis bending efficiency to the box, lower mass, but much weaker torsion. Eccentric cam/follower forces need anti-roll shoes.
T section	Unsymmetric centroid and shear center; weaker lateral restraint, twist under eccentric load. Useful as a guided rib, not an unsupported replacement for the box.
Warren / triangular web	Chords separated in height provide bending stiffness; triangulation transfers shear. Chord-only EI is an **upper bound** because diagonals stretch and joints rotate. Compression diagonals need buckling checks; thin FDM nodes concentrate stress. A plane truss is torsionally weak.
Pratt	For one load direction, long diagonals can mainly carry tension and short verticals compression. Selection/withdrawal can reverse forces, so the favorable load sign is not guaranteed. More nodes and a weaker reverse-load case remove much of its advantage at this scale.

Structure	Bending / torsion / manufacturing result
Ribbed/lattice box	Keep closed outer skins and add ribs at cam-force entry/support points. A sparse internal lattice helps local crushing; it adds little global EI unless it transfers shear between separated chords. Decorative triangles do not create torsional closure.
Two insertedØ2-mm CF rods	Put rods near opposite vertical extremes and transfer shear through the printed core. Merely pushing loose rods into bores does **not** earn the parallel-axis composite-EI term. The model gives both unbonded/sliding and full-composite bounds.
One continuous stock rod + printed attachments	Feasible when shoes carry alignment and anti-roll. A round rod alone does not set attachment angle or cell pitch. It trades less printing for many bonded/clamped attachments and a joint-slip tolerance stack.

The selected printed core is a box with **twoØ2.2-mm channels for nominalØ2-mm rods**, centered at z 2.2 and 8.1 mm. A 0.8-mm internal vertical web supports the upper rod sleeve during printing and carries local shear. Its extra web, channel sleeves and cam rails make it heavier than the idealized bare box table; the actual CAD volumes are in the mesh manifest. The table intentionally does not count a cam as a full unnotched chord. The main lower box is uncut by the repeating follower openings. A separate shared upper cam cap allows followers to be placed without threading every small part through a closed track. This cap is a repeated **row component**, not another per-cell part.

A local cam lip is a short cantilever, not the whole selector. With 0.6-mm projection, about 0.7-mm thickness and 0.8-mm loaded width, 0.75 N on one lip gives order 0.002-mm elastic movement and 7 MPa nominal bending stress atE2 GPa. Corner stress, layer adhesion and contact crushing are omitted. The thin lip and the cap's withdrawal load therefore remain physical inspection points even when the main beam is stiff.

Stock-material comparison

Commercial dimensions/prices and exact source links are in sources.md. Mechanical properties not stated by a source are explicit screening assumptions.

Stock	406.4-mm nominal mass	Practical implication
CFØ2 rod, density 1.5	1.92 g	Very light; at 25.4-mm supports, low-end 20-GPa bending modulus gives roughly 0.068 +0.065 mm in the normalized cases. One tiny rod is not a 400-mm bridge.
CFØ3 rod	4.31 g	More efficient as a mostly purchased member; about 0.0135 +0.0129 mm at 25.4 mm. Attachment angle still needs shoes.
CFØ3/2 tube	2.39 g	Better bending per mass than a 2-mm solid; thin wall is easier to crush at a clamp.
GlassØ3 rod, assumedE30 GPa	5.75 g	Cheap, insulating and forgiving; stiffness must be measured for the bought variant.
SteelØ2 rod	10.0 g	Cheap and stiff;≈0.0068 +0.0065 mm at 25.4 mm. Higher moving mass but much less axial shortening.
SteelØ3 rod	22.6 g	Mechanically simplest stiff stock option. Less printing with attached cam strips; heavier than CF but inexpensive.
AluminiumØ3/2.1 tube	3.96 g	Useful compromise;≈0.0051 +0.0049 mm at 25.4 mm. Avoid point set-screws on the thin wall.

Do not use generic aerospace carbon modulus. The linked manufacturer's TDS lists tensile 28–40, flexural 20–30 and compressive 10–20 GPa. Calculations use the lower 20-GPa bending value and 28-GPa tensile value for the reinforced member's pull stroke. A pushed CF member can shorten substantially more. The pure-stock CSV's EA uses its listed screening E; use the TDS's appropriate tensile/compressive modulus for an actual actuator direction.

Recommended construction

Use the **guided closed printed cam beam with continuous inserted reinforcement**, assembled from four approximately 101.6-mm print sections along each full row. Two 2-mm steel rods are the conservative low-cost default at the calculated worst row force. Two 2-mm CF rods are the lighter alternative, conditional on measured axial stiffness and joint slip. The same printed channels accept either material. Use paired guide shoes at **25.4-mm intervals initially**. Their placement and attachment to the stationary guide structure are part of the load path. Do not infer stiffness from the 4-mm bore alone or leave a long span unsupported.

Pull the loaded ON stroke from the far end so the CF primarily carries tensile load. Local printed sections must transfer axial cam load into the stock: use prepared, bonded sleeves or positive end capture plus a verified continuous load-transfer path. For a roughly 100 N worst-case row force, each of two rods carries order 50 N. A25-mm bond length on a 2-mm rod has nominal average shear about 0.32 MPa, but peel, surface preparation, adhesive creep and print-layer failure can dominate. Adhesive is not assigned a guaranteed strength. Do not cut screw threads into carbon or use a crushing point clamp. Use broad split collars/end shoes; leave the far support free to accommodate length change.

A long printed shell can be thermally incompatible with a bonded rod. Using illustrativeα values PLA70, aluminium 23, steel 12 and axial CF0–3 µm/(m K), a 20 K change over 406.4 mm creates about 0.54–0.57 mm PLA/CF mismatch. These α values are assumptions. Short printed attachments with controlled gaps reduce accumulated thermal stress; tightly bonding the entire length gives better composite stiffness but more thermal/creep risk. The short-support design deliberately does not require the ideal full-composite EI to succeed.

Check supplied rod straightness on a flat reference and dry-fit all shoes. The seller's OD tolerance is not a straightness specification. Shoes should constrain roll/lateral position without forcing a bowed rod into high friction. Rod-to-sleeve clearance, joints, bond slip and shoe clearance belong in the same measured stroke budget. A single rod with tiny individual clips needs 6,400 angular/pitch placements; longer printed multi-cell cam strips reduce this assembly burden.

Column roofs are unloaded when changed in the normal sequence (rows returned first). Their split-rail section must still resist the transverse load of blocked half-selected followers while stationary. At 25.4-mm supports, two approximately 1.1×6.5-mm rails give an estimated 0.02–0.04-mm combined normalized distributed/point deformation at E=2 GPa (roughly twice that at E=1 GPa), before guide flexibility. They do not need the same tensile reinforcement as the loaded row pull. Attempting a fully loaded column return across 80 cells is a different force case and is not credited as a qualified low-force operation.

Deformation allowance and actuator force

The cam has a **0.30-mm dwell half-width**, and the nominal local spring compression after contact is 0.8 mm. An initial 0.15-mm budget for beam movement plus connection error consumes half the dwell margin and changes nominal spring force by about 0.21 N. This is a functional tenth-of-a-millimeter requirement, not a micrometer positioning demand. It excludes measured rod bow and frame misalignment until those are characterized; all cannot each consume the whole budget.

For cam slope $m=1.4/(2.54-0.6)=0.722$ and assumed$\mu=0.2$, input-force ratio is $(m+\mu)/(1-\mu m)\approx 1.078$, plus shoe friction. At 1.102 N/cell, 80 loaded cells require about **113 N** during the ramp. At 1.79 N/cell the same assumptions give about **183 N**. The 0.3–1.79 N and µ0.10–0.35 sweep is supplied. Dwells remove the ideal geometric holding force; they do not remove friction, tolerance or motor retention requirements. A 5:1 edge mechanism would trade about 12.7 mm actuator travel for lower force, before its losses. It is an actuator interface allowance, not a designed motor bank.

A bare printed box can shorten about 0.49 mm under the normalized 100 N distributed axial load over 406.4 mm. Two properly loaded 2-mm CF rods reduce the tensile estimate to about 0.093 mm; bond slip could erase that gain. This axial result is a stronger reason for reinforcement than the already small transverse deflection between shoes. Steel is an economical fallback where moving mass is acceptable.

One structural measurement, not another print campaign

The same plate includes a 100-mm representative beam and two supports. Compare empty channels with inserted rods on 50/75-mm support spans using the **same** beam, same load and orientation. Record rod material/diameter, measured span, load and midspan movement. Loose rods measure the unbonded case. They cannot validate full-composite action; if the owner elects to bond the same specimen, record that separately. Do not infer 406-mm

performance from a short beam without preserving its support/connection model.

The final bed-supported follower geometry (0.40-mm thickness, 3.6-mm depth, 0.9-mm vertical leg pitch) predicts 1.102 N selected and 1.79 N half-selected at E=2 GPa. The final force CSV includes these values: about 113 and 183 N per 80-cell row at μ=.20. Scale the distributed deflection linearly to 1.79 N/cell; the 3-N single obstruction remains a separate case.

The reproducible Warren-web screen uses 5.08-mm panels, 8.4-mm chord separation and a 2×0.8-mm diagonal. Its effective shear rigidity is GA≈EA_diagonal a h^2/l^3. Over 25.4 mm, web shear alone adds about 0.013 mm under the normalized distributed load and 0.016 mm under the 3-N point load, **on top of** the chord-only bending entry. The combined maximum diagonal force is about 4.7 N versus roughly 17.5 N ideal pinned Euler buckling capacity. Thin-joint rotation, printed defects and reversed load reduce that margin. This is why a space-filling triangle pattern is not automatically superior to a closed box at this pitch. Exact values are in analysis/results/truss-web.json.

At the final 183-N half-selection row case, the lower-bound CF tensile model predicts about 0.17-mm distributed axial shortening, slightly above the initial 0.15-mm budget before joint error. The corresponding two 2-mm steel rods in the same box predict about 0.029 mm. Steel adds roughly 16 g per row versus the carbon rods (about 1.3 kg across 80 rows), but this extra inertia is negligible beside the hundred-newton cam force. Eighty 1-m steel rods cost about€80 including listed VAT versus€156 ex VAT for the carbon tier. Thus **steel is the simplest conservative reinforcement for this force envelope**; carbon remains attractive if its measured modulus/joints and allowed stroke error justify the weight saving. No stock is bought or separately prototyped here.

E-304 — guided cone, captured row cam, column ceiling

Scope: one concrete bench experiment, not a replacement for DES-009. All performance below is calculated or geometric unless explicitly attributed to owner testing in evidence-and-selection.md. No E-304 physical result exists.

Selected arrangement and mechanical AND

Each resident screw has a downward-facing 30° half-angle female cone. An always-meshed input gear and male cone share one sliding shaft. Two stationary journals locate that shaft; two separate stationary journals locate the output screw. A nonrotating follower presses the input through a small central thrust contact. A retaining lip withdraws it. Neither moving selector locates a rotating shaft.

A full-length row slides **2.54 mm along its length**. Its captured cam track lifts the follower bottom 1.4 mm. A perpendicular full-length column slides the same distance, changing the follower's upper stop. The stop is 0.1 mm above OFF in the low state and 1.1 mm above OFF in the high state. The cone seats after nominally 0.6 mm of shaft travel. The remaining 0.8 mm compresses the follower. This produces the following truth table:

Row	Column	Input lift	Follower compression	Contact
0	0	0	0	Open
0	1	0	0	Open
1	0	0.1 mm	1.3 mm	Open; ceiling carries spring load
1	1	0.6 mm	0.8 mm	Cone loaded

The 0.3-mm flat dwell at each end of the cam makes state retention less sensitive to end-stop positioning. The 1.94-mm sloping run gives a steep 0.722 rise/run. It is easy to manufacture but does **not** provide large force reduction. In normal operation unload rows before moving columns; this follows from the force calculation, not a need to search rotational phase. Loaded column release is included in the bench observations and must not be forced.

The compliant feature is justified here by an explicit conflict: a rigid row must finish its stroke when one column blocks a follower and another cone seats early. A rigid connection cannot satisfy different local stopping heights. The six-fold printed flexure supplies that accommodation without a separate spring, holder and assembly operation. It is a spring function with creep and fatigue risk, not complexity-free plastic. Other architectures may solve the conflict differently.

Dimensions, constraints and force paths

The exact E-295 thread and pin meshes are retained from E-297's archived copies: Ø3.2-mm screw, 3-mm lead, 4.4-mm square pin, approximately 6-mm internal threaded region. Thread mesh is translated to Z55–71 mm; the pin is at Z61–79 mm at neutral. Positive screw rotation gives **negative** pin translation for this actual handedness. The bench range is ±1.5 mm, not a full-height terrain stroke.

Function	Delivered geometry	Consequence
Input radial/tilt support	Ø1.8 shaft / Ø2.06 guides; centres Z33.6 and 38.6, span 5 mm	Two stationary lands; nominal diametral clearance 0.26 mm
Output radial/tilt support	Ø1.6 neck / Ø1.88 guides; centres 46.6 and 51.8, span 5.2 mm	Screw location independent of row/column deflection
Output axial support	Ø2.8 collar between lower journal top Z52.4 and upper keeper Z53.4; collar thickness .8 mm	.2-mm axial float; loaded cone pushes collar against the upper deck; gravity can place it on the lower deck
Input axial force	Central dome/pad, effective friction radius screened 0.267 mm	Lower drag than E-300's rotating contact near r 1.85 mm

Function	Delivered geometry	Consequence
Release	Follower cage around rotor flange, 0.3-mm running clearance	Positive withdrawal after lost motion; not gravity-only
Terrain antirotation	4.7-mm square guides at 64.5 and 75.8	0.3-mm nominal total square clearance; both guide planes remain engaged over bench stroke
Cone	r 1.1–1.9 mm; cup OD4.6, half-angle 30°	Neighbor cup gap 0.48 mm; axisymmetric ideal contact
Permanent drive	Module 0.3, 12-tooth, 25° pressure angle; OD4.2 mm	Neighbor gear gap 0.88 mm; no selection-time gear meshing
Row structure	4.6-mm width, 10.4-mm box core plus cam walls to 14.7	0.48-mm neighbor gap; two Ø2.2 stock channels
Column	Two parallel 1.1-mm rails; minimum roof depth 5.5 mm	Central slot clears follower; fixed transverse guides needed
Pin neighbor clearance	5.08−4.4=0.68 mm	Actual retained pin, not an enlarged stand-in

Axial load goes row cam → follower spring → central thrust contact → input cone → output cone → collar → stationary deck → frame → row guide. Torque goes common rack → permanent input gear → cone surface → output neck → retained screw thread → pin. Radial cone and gear reactions go to fixed journals. Columns carry the spring reaction at half-selected cells. Captured row tracks keep the follower bottom against the cam during release; the upper stem and feet have shared stationary guides.

The 2×2 frame opens on opposite sides of its two rows. Two shared keeper strips per row close the guide slots. A separate cam cap allows followers to be placed into their tracks rather than requiring threading every follower through a closed comb. These are real additional shared parts. The bench uses tape at external tabs for retention. A large array would need shared split guide decks/caps with accessible edge or top assembly and retained joins; the complete large-array assembly/service route is **not yet designed or qualified**. A fully inaccessible interior-pixel repair is unacceptable and remains a scale-up gate.

The mechanism consumes about 55 mm below the working thread and 79 mm to the neutral pin top in this coupon. The reinforced row costs roughly 13 mm more depth than a flat 1.6-mm strip. That is plausible for a tabletop base but a significant height penalty. No project evidence establishes that this is an accepted final enclosure height. A shorter optimized stack is deliberately not another prototype here.

Force, torque and tolerance budget

The six-leg follower is analysed as a plane Euler–Bernoulli beam frame, constrained in lateral translation and end rotation. At nominal E=2 GPa, width 3.6 mm and thickness 0.40 mm, k=1.378 N/mm. At 0.8-mm compression F=1.102 N and peak linear bending stress≈19.1 MPa; half-selection at 1.3 mm gives 1.79 N and≈31.0 MPa. Curved joints, large displacement, print anisotropy, residual strain, contact and creep are not resolved. E=1.2–2.8 GPa and t=0.35/0.40/0.45 are tabulated. Vertical leg pitch is 0.9 mm; the lower neck clears the row tracks and connects the bed-supported foot and spring across a short printed bridge. The flat back is at Y=−1.8 mm so the six ligaments print directly on the bed. A small back web supports the column foot. The two thickness alternatives are diagnostic coupons; four nominal followers test simultaneous behavior. Do not keep half-selected followers compressed in storage.

For a cone under uniform wear, gross torque is $T=\mu_c F R/\sin\alpha$, with R=1.5 mm and α=30°. The equivalent coefficient is $3\mu_c$ Nmm/N. Unlike a flat disc, the cone increases normal force and can centre within guide clearances. It also introduces radial forces and possible wedging. The simple release screen $\mu_c<\tan 30°=0.577$ is necessary context, not proof against local staircase interlocking or adhesion.

A conservative working calculation subtracts input/output thrust drag and an explicit unknown lateral term:

$T_{net} = F\,[3\,\mu_c - 1.437\,\mu_t - k_{lateral}]$ Nmm, with $k_{lateral}$ screened 0–0.25 Nmm/N.

At μ_c=.25, μ_t=.20, $k_{lateral}$=.10, F=1.102 N, **T_{net}=.400 Nmm**. This exceeds E-303's assumed light starting torque .0597 Nmm and only narrowly exceeds its assumed moderate .343 Nmm. At μ_c=.30 the same model gives .565 Nmm; at μ_c=.15 only .0690 Nmm. Thus neither moderate torque nor even reliable light margin is established by

CAD. Raising preload also increases half-selection force, creep and guide losses. E-303's loaded/imperfect 2.20 Nmm and obstruction 9.80 Nmm cases are outside this small clutch's intended capacity; slipping is preferable to forcing a blocked pin.

The cone contact area is approximately 15.1 mm². At F=1.102 N, total normal force 2.20 N gives a nominal average pressure≈.146 MPa. FDM ridges initially carry only part of that area; local contact stress can be much higher. Bidirectional friction capacity is symmetric in the ideal model. Reversal backlash comes mainly from the permanent rack and retained screw, and surface wear can change both grip and release. Slow manual testing controls heating; continuous motor duty is unqualified.

At 1 N/cell and μcam=.20, an 80-cell row needs about 102 N during its steep ramp, including an estimated sliding shoe term. At the nominal 1.102 N/cell this is≈113 N; eighty half-selected cells at 1.79 N approach 183 N. A two-cell bench stroke should be a few newtons. The load is mainly spring/cam reaction, not selector mass. Column gate friction may reach μ·ΣF (about 29 N at 80 half-selected cells) before roof-ramp effects. This is why columns normally change with rows unloaded. No claim is made that a cheap micro-servo can actuate a complete row.

Differential row/column deflection changes local compression rather than moving shaft centres. A±.15-mm height error changes nominal force by±.207 N. That allowance is related to the 0.8-mm preload and 0.3-mm cam dwell, not an arbitrary precision requirement. A blocked follower compresses further, so nearby followers can still travel **if** the selector completes its stroke. The actuator and support must still withstand the extra force; compliance cannot eliminate total reaction.

Tolerance deserves separate caution. Input radial clearance is 0.13 mm and output 0.14 mm. Opposing radial offsets can total 0.27 mm before tilt. Individual worst-case tilt screens are roughly 0.26/5=3° and 0.28/5.2=3.1°. Their simultaneous extremes are not an operating prediction, but can consume the 0.6-mm axial opening gap. The supplied actual-mesh phase/eccentricity/tilt sweep reports first contact rather than silently assuming coaxiality. Fixed guides bound wobble; they do not prove freedom from rubbing. Nominal,−.10 and+.10-mm input shaft variants test this fit tradeoff. Tightening guides without print evidence is not a solution by assertion.

Complete repeated mechanism and scale costs

Item/function	A1 guided cone, selected	Two individually guided cones in series	Guided dog plus local lost motion
Printed moving bodies per pin, including terrain pin	4	8	5
Purchased local parts	0	0 in all-printed version	0 with integral compliance
Independent local elastic functions	1	2	1
Rotating bodies	2	3	3
Radial support planes	4	At least 6	At least 4 plus sliding spline guidance
Separate local thrust/release interfaces	Central input pad, output collar, withdrawal lip	Two clutch reactions, intermediary thrust/release	Axial dog reaction, follower and spline interfaces
Pin guide planes	2	2	2
Components for 6400 pins before shared hardware	25,600	51,200	32,000
Main residual risk	Friction margin, guide rubbing, flexure creep	Extra drag, stack height, doubled adjustment	Angular engagement conditions and spline friction

The four A1 bodies are pin, output-screw/cup, input-gear/cone and follower. They contain three rigid motion functions (pin translation, output rotation, input rotation+axial sliding) plus follower deformation/translation. Count the input's two degrees of freedom and the six bending ligaments explicitly; integrating them into four STLs does not remove those functions. The stationary frame supplies repeated sliding-bearing surfaces without 6400 separate bearing inserts. Each cell still needs at least four part placements, threading the pin, and checking free motion, grip and release. At 10–30 seconds per cell that alone is 18–53 hours for 6400 cells, excluding shared assembly; this is an

illustrative labour sensitivity, not measured work time.

At 80×80, long functional members are 80 row bases+80 caps+80 columns+80 drive racks+160 keeper strips=480. Add two common drive/bus functions, fixed-guide components G, edge actuators and retention hardware. Ideal continuous-part count is therefore 26,082+G before edge hardware. Four printed sections per long member would instead give 27,522+G before joins/hardware. This segmentation is **passive manufacture of one continuous selector**, not separately controlled modules. End-to-end alignment and cap retention need evidence. Two 2-mm carbon rods per row require 160 cut lengths, about 430 mm including edge attachments:80 one-metre stock rods if two lengths can be cut from each. Rod price is€156 ex VAT at the observed 20+ tier, or€244 at single-unit prices; cuts and shipping are extra. Columns initially use printed rails with fixed supports; adding their reinforcement would increase cost.

Full-array budget line	Illustrative allowance	Basis / limits
Printed material	€120–300	6–12 kg at€20–25/kg; covers cells, large shared guides, waste; not a sliced full-array BOM
Row reinforcement	Steel≈€80 incl listed VAT; CF€156–244 ex VAT	80×1m rods, two cuts each; steel is conservative for the final 183-N force case
Common bearings, joins, retainers, drive stock	€75–250	Budget allowance; common-drive and guide assembly remain unresolved
160 selector actuators/transmissions	€800–2,400	€5–15/axis placeholder, **not a vendor quote or force-qualified actuator**
Shared motor/transmission	€50–200	Force/duty dependent placeholder
Power, drivers, wiring	€100–400	Architecture-dependent allowance
Total excluding labour, shipping and tax normalization	**about€1,300–3,800**	Broad screening range; not a purchase-ready cost claim

The source-price table uses mixed tax bases explicitly. The actual prototype mesh volume and print usage are separately recorded; do not infer a complete manufacturing quote from four cells. The two-series design roughly doubles the local mechanism parts, adds at least one more rotating assembly and stationary support plane pair, and raises repeated material/assembly cost. It could still win if single cones can use fewer stressed interfaces, but no evidence currently offsets those penalties. Printed-only supported row boxes save the corresponding rod cost; they merit the included short beam comparison before any purchase.

Shared power and what 5.08-mm feasibility means

The bench has two permanent racks joined to one shuttle and one 36-tooth hand pinion. It drives all four input rotors in the same direction, even when no clutch is selected. No individual hand-positioned input substitutes for common-drive engagement. Each 10° knob notch corresponds to 30° input rotation and 0.25-mm ideal pin travel. The larger side pinion is outside the raster.

For an 80-wide field the same principle is a full-length rack per row, all coupled to one bus. At moderate assumed screw torque .343 Nmm, load-only rack force is Nselected×.343/1.8:15.2 N for 80 loaded pins, 1.22 kN for 6400. The corresponding 5.4-mm pinion torque is .082 and 6.59 Nm before drive and clutch losses. All unselected input bearings still rotate; their drag is unmeasured. These numbers prevent an unjustified claim that one tiny motor solves the array. No update-scheduling research is repeated.

A40-mm pin stroke with 3-mm lead requires 83.8 radians of screw rotation and 150.8 mm of rack travel; a rack covering the 406.4-mm field would need roughly 557 mm plus end margins. The bench provides a short travel representative common drive, with a marked rack window−9.5 to+2.8 mm about the CAD assembly position. Full phase testing starts at−8.5 mm and resets with clutches OFF; the input cannot rotate indefinitely. A different continuous common transmission is possible, but is not designed or claimed here.

Local packing is plausible: actual retained threads, four neighboring cups/gears, two guide spans, shared selectors and fixed decks fit at 5.08-mm pitch in the exported assembly. **The complete 80×80 mechanism is not yet demonstrated manufacturable or economically superior:** full-stroke common power, large guide-deck assembly, thermal/print variation, selector force, repair and lifetime remain unresolved. The only next evidence sought is the

small physical result defined in the owner pack. Development stops for an explicit owner decision afterward.

Same-assumption comparison of the strongest candidates

For a common screening target 0.343 Nmm, μc=.25, μt=.20 and lateral allowance .10 Nmm per newton, the single 30° cone requires F≥.946 N per cell. The chosen nominal linear spring prediction is 1.102 N. The corresponding flat annulus has coefficient 1.5μc−1.437μt−.10=−.0124 Nmm/N: under this adverse drag assumption **more preload cannot create a useful net margin**. At μt=.10 it instead needs 2.61 N/cell; it remains a real candidate if measured thrust drag is low. “Half the gross torque” must not be mistaken for merely double the net force.

Two identical cones in series require both stages to exceed the target, not half the target each. If the extra intermediate assembly contributes a separately assumed .05 Nmm drag, each stage would need at least 1.08 N in the same screen; actual force placement changes that result. Each full row/column then directly operates 80 preload functions, approximately 110 N/ramp at the present cam ratio. A has one controlled normal force, with its highest approximately 183-N row case arising from blocked half selections; B removes the roof-gate contact but adds the second preload, intermediary journals and assembly.

A two-dog clutch at r 1.5 mm transmits .343 Nmm with total tangential reaction .229 N. Even two .24-mm² nominal load faces would have only .48-MPa average shear/bearing stress. That attractive torque scale does **not** answer whether the teeth seat. Its minimum clearance and engagement force depend on the keyed sleeve and chosen indexing profile. The repository's roughly .3-mm double-D diametral allowance did not prevent wobble; A removes that fit, B and the selected A3 concept retain at least one extra sliding torque fit. A3 therefore remains unranked for reliable stationary-phase operation rather than being assigned a fictitious success probability.

All candidates can use the same full-length supported beam. None earns an unsupported 400-mm span or automatically cheap actuation. B adds at least 6,400 intermediate shaft assemblies and 6,400 more compliant followers compared with A, before sleeve parts; guide surfaces and friction scale with them. A2's component count/cost matches A but its pressure or bearing design changes. A3 has at least 6,400 additional sleeves and repeated sliding connections. For all three, every local fit would need inspection and replacement access; only A receives CAD and one print in this assignment.

Explicit adverse guide-clearance corner

At cone height Z42, an input journal centre at +.13 mm in its upper guide and−.13 mm in the lower guide extrapolates to+.307 mm. Opposite output offsets (lower−.14, upper+.14) extrapolate to−.388 mm. The relative cone eccentricity can therefore approach .695 mm in this deliberately adverse rigid-clearance corner, exceeding the nominal .6 tan 30°=.346-mm radial opening allowance. The mesh sweep includes .4-mm/3° and .7-mm/6° imposed cases to expose the resulting OFF interference. **The delivered tolerances do not guarantee drag-free disengagement at every possible guide-clearance corner.** Cone centring under load and ordinary gravity/rack forces may prevent that corner in practice, but this is not established. The fit variants and the OFF shared-drive test are essential; an owner result showing rubbing is a no-go for this exact fit, not something to “solve” by forcing or phase searching. A production clearance/guide-length revision is deliberately left to a decision after the physical result.

The output collar has 0.2-mm axial clearance between its two stationary faces. The nominal loaded CAD places it at the upper face. Under gravity it may fall 0.2 mm; ideal coaxial initial contact then begins at input lift 0.4 mm and final loading lifts the output back to its upper stop. This small terrain-height jog is observable and must be recorded. With the column OFF and input at 0.1 mm, the remaining nominal gap can be 0.3 mm. Combining axial float with adverse radial offset further reduces OFF clearance: the .27-mm offset sweep's roughly .13-mm first-contact lift becomes roughly−.07 mm after a .2-mm output drop. The broad guide tolerances therefore need physical fit evidence; nominal no-collision results are expressly not a worst-case tolerance qualification.

At .343 Nmm, the solid 1.6-mm output neck has nominal torsional stress $16T/(\pi d^3)\approx.43$ MPa; the thin teeth, printed interfaces and guide fit are more credible first failures than bulk shaft torsion at this load. A12-tooth 25° full-depth involute is above the ideal unshifted undercut screen $2/\sin^2 25°\approx11.2$ teeth. This geometric result does not guarantee printable tooth tips: the .3-module pitch thickness after backlash is about .39 mm and is explicitly checked in the Arachne toolpaths. A .12-mm layer on a 30° cone produces an ideal radial staircase step of about .069 mm, so the smooth-cone friction model is only a starting approximation.

Physical component total, including stock and edge functions

The preceding 26,082+G figure counts printed parts/functions before stock and actuators. Add 160 cut reinforcing rods and 161 actuator assemblies (160 selector axes plus one shared rotary source in the explicit direct-actuation budget): **26,403+G+H total component assemblies**, or **27,843+G+H** with four printed sections per long member. H denotes unresolved fasteners, rod-end retainers, segment joins and electrical hardware; motor internals are not disassembled into fictitious individual-part counts. G is explicitly unknown because the full stationary guide structure is not designed. An exact finished-array total cannot honestly be given at this stage. The complete bench assembly contains 29 retained printed parts, plus ordinary retention tape; the extra four fit/preload variants and three structural-test pieces are not installed simultaneously.

For B, replacing the column roof with a second captured cam also adds 80 column caps, so a comparable shared printed count is 560 long functions rather than 480. Its continuous printed subtotal becomes 51,762+G_B. If both loaded selector sets receive two rods per selector, add 320 cut rods and the same 161 actuator assemblies: **52,243+G_B+H_B**, before sectioning. This explicitly includes the intermediate rotating assembly in the eight local bodies. Unresolved guide/retainer counts are retained as variables rather than hidden zeros.

Unselected height retention is also unproved

No per-pixel holding brake is hidden in the four-part count. With lead 3 mm and mean diameter 2.3 mm, the simple 90° V-thread self-locking threshold is approximately $\mu_{thread} \geq [3/(\pi \cdot 2.3)] \cos 45° \approx .294$. E-303's light running scenario predicts a small negative lowering torque (about−.0021 Nmm after its assumed guide/thrust/drag terms), so unselected loaded pins may backdrive. The owner's smooth hand-fit result did not measure retention. The bench therefore observes unloaded and lightly loaded pins with every clutch OFF before crediting isolation. Rapid gravity-driven descent is a no-go; small drift is recorded according to the project's present priorities. Adding a brake would change repeated components and cost and is not assumed or developed in this assignment.

E-304 verification and readiness

Print status: DRAFT. Do not treat the supplied 3MF as a READY print instruction. The archive is a genuine configured Bambu Studio X1 Carbon project and has been reopened/resliced. The remaining blocker is its critical deposited geometry: the retained pin wall has a maximum nominal midwall discontinuity of **0.207 mm**; layers **12.48, 15.48 mm** still fail the closed-loop screen even after an 0.08-mm geometric buffer. This is a slicer finding, not an observed broken print. A continuous watertight STL does not establish continuous deposited walls.

The operator has not physically printed or assembled E-304. No failed or successful owner result is invented. The owner's requested DRAFT fallback and owner-print-tests/AGENTS.md apply: successful saving/reopening alone does not establish print readiness. Existing READY offers and DES-009 remain unchanged. Stop here for the owner decision; no further design or print campaign is initiated.

Calculated and geometric evidence

-15 structural section alternatives, 216 torque cases, 54 flexure cases, 1,440 representative tooth phases. The E-303 screw function is called directly with its original light/moderate assumptions.

- Independent single-member fixed-guided beam limit agrees with the frame-element solver; actual box/web/sleeve area and second moments are sampled at 0.01-mm grid spacing. These are numerical checks, not material measurements.

- Delivered meshes are watertight. Four supported screws deliberately form four bodies in one named print object. The frame's ten sacrificial rods are explicitly removed from the operating assembly.

- **32 nominal assembly configurations**:16 row/column masks, six sampled intermediate selector strokes, four common-drive positions in both directions, one blocked follower and one stalled output, plus four rack-end configurations (OFF and selected with compatible independent output phases). No unexpected nominal mesh interference above 0.001 mm³ was found. Matching cone surfaces show 0.004–0.0073 mm³ tessellation overlap at the nominal mathematical contact position; these are explicitly labelled contact, not silently discarded collisions.

-25 common-rack positions through one complete tooth pitch have no unexpected gear/frame intersections. A deliberately displaced rotor produces a detected interference, checking that the screen can fail. A separate+4.5-mm rack-overtravel case detects approximately0.070 mm³ interference between the shared shuttle crosspiece and hand pinion. The operating window is therefore restricted to−9.5…+2.8 mm, with the phase sweep beginning at−8.5 mm. The bench uses marked manual limits; unlimited knob rotation is not allowed.

- Actual faceted cone contact is sampled at 72 phases for nine offset/tilt cases (648 cases). Coaxial first contact varies about 0.596–0.597 mm due tessellation, versus 0.600 mm for exact ideal surfaces. At .1/.2/.27-mm imposed radial offset first contact advances to about .425/.252/.131 mm. Deliberately adverse .4-mm/3° and .7-mm/6° cases **interfere while OFF**. See the CSV and the explicit journal-clearance/axial-float corner calculation; these are not tolerance-qualified operating states.

- The actual thread handedness is retained: pin height changes−3 mm per positive screw revolution. The bench's±1.5-mm travel, phase offsets and both output guide planes are included. No full 80×80 model or full-stroke drive is claimed.

Actual saved project

Bambu Studio **02.08.02.61**, official X1 Carbon 0.4-mm printer/machine-G-code preset, Generic PLA and process settings inherited from the physically tested E-295 compact-thread project, with the explicitly recorded E-304 process changes. This was an actual Bambu **command-line** save/reopen/reslice, not a claimed manual GUI action. No external printer/process/filament settings were loaded during reopen. Saved settings compare equal. The archive contains 33 named objects and actual G-code; it is not a mesh-only 3MF.

Printer: X1 Carbon 0.4 mm; PLA220°C, textured PEI55°C; layers/first layer .12 mm; Arachne, two walls, 100% zig-zag infill, nominal line widths .45 mm, minimum bead 75%, minimum feature 10%. Outer/inner speed 15/20 mm/s, first layer 15, bridge 15; general acceleration 1000, outer/first 500 mm/s². Supports OFF, raft 0, outer brim 2 mm with .12-mm gap, by-layer order. XY/hole/elephant-foot compensation 0. Fan 100% after first layer, auxiliary fan 0, cooling minimum speed 10 mm/s. Bridge direction 90°, seam gap 0%. Full values and source hashes are retained in JSON.

Original slicer estimate:7 h 34 min 27 s /21.60 g PLA. Reopened estimate:7 h 34 min 17 s /21.60 g. These are slicer estimates, not physical consumption or time. The combined one-plate test is small in material, but deliberately slow for fine threads, gears and thin flexures.

Archive geometry matches the bed-oriented source plate to a maximum bidirectional vertex error of 0.00002254 mm, with equal facet counts. This is a coordinate fidelity check, not a printer accuracy specification. All 33 objects have deposited paths through their interior height. Source orientations, supports and brims are saved rather than left to automatic placement.

Critical toolpath review and remaining blockers

The focused path figure and full atlas are drawn from the actual archived G-code. Model-path analysis excludes brim/skirt, so brim is not incorrectly credited as part of a structural wall.

- The six follower ligaments print on their flat back. The short lower-neck bridge is about 2.4 mm. Clearance between folded ligaments remains visible; integral compliance is not physically calibrated.

- The permanent gear has 12 deposited teeth. This establishes representation of the tooth geometry, not tooth strength, backlash or a successful mesh in plastic.

- Both rod-channel roofs and box roofs have transverse bridge paths after specifying the 90° direction; the upper rod sleeve has a supporting internal web. The 100-mm beam no longer starts a large detached upper-sleeve island.

- The frame has no fully isolated new deposited components in the screened layers after adding sacrificial supports. Its largest screened remaining free overhang/bridge centreline is about 16.2 mm. This still needs real print evidence; the screen is not a structural guarantee for a thin bridge.

- Small isolated end beads remain on the row base and cap, approximately 0.15–0.16 mm² each. They are at inactive outer ends, not the two active coupling positions, but they are recorded rather than called a perfect slice. Cleanup and cap retention remain untested.

- Screw scaffolds appear as connected paths at their selected tab levels; no whole screw/rotor/follower component begins unsupported in the all-layer footprint screen. Actual tab removal, scar depth and adhesion remain unknown.

- The pin bore stays open, with minimum nominal axis-to-material clearance about 0.839 mm. Nevertheless, 22 of the sampled pin layers lack an enclosed footprint loop and two critical layers retain roughly 0.207-mm outer-midwall gaps. Setting seam gap to zero did not remove those two gaps. **This prevents a READY claim.** No further profile or thread modification is hidden in the delivered file.

Original and reopened G-code are not byte-identical. Independent comparison finds a maximum per-layer nominal deposited-footprint difference of **1.227%**. Segment counts/order and small geometric choices differ; both archives receive separate object/path reports. This is reported as a limit rather than presented as exact toolpath reproducibility.

Repository validation and delivery integrity

Python sources compile. Model checks, mesh/phase reviews, both Bambu slices, ZIP integrity and source/archive coordinate comparison were run. New relative documentation links are checked. Every pre-existing tracked file, including DES-009 and historical owner results, is unchanged.

./repo check exits nonzero on six **pre-existing** size-policy findings: atomic objects E-266, E-283, E-286, E-288 and E-293, plus owner-print-tests/AGENTS.md. Running the same check in a clean worktree at baseline 1071f 695b 93951246a 2865038e 7fb 9de 9bdf 2a 05 yields the identical six findings. They are not fixed in this unrelated task. No new E-304 structure/reference finding is introduced.

The experiment is delivered on a review branch and draft pull request. Nothing is merged into main, and no physical validation or further development is authorized by producing the files.

Sources and provenance

Read 25 October 2026. Prices are observed listings, not quotations or authorizations to buy. VAT and quantity bases are retained; currencies are not silently converted. Availability and delivery can change.

Source	Extract used and limitation
Easy Composites EU, Ø2-mm CF rod	1/2-m stock; €3.05/m ex VAT single, €2.60 at 5–19, €1.95 at 20+; listed OD tolerance 0.1 mm, interpretation not specified as ±. No straightness guarantee found.
Manufacturer pultrusion TDS v 1.0, 20 September 2024	Typical tensile modulus 28–40 GPa, flexural 20–30, compressive 10–20; density 1.30–1.50 g/cm³. Use the relevant modulus, not an assumed generic 120–150 GPa. These are typicals, not lot acceptance limits.
Easy Composites EU, Ø3/2-mm CF tube	1/2-m stock, €2.80/m ex VAT single, €1.85 at 20+; OD and ID tolerance each listed 0.1 mm. Thin wall needs distributed clamps.
Easy Composites EU rod range	Ø0.5–12 mm plus 4-mm square stock; Ø3-mm round from €3.15/m and 4-mm square from €4.55/m are quantity-dependent “from” prices, not single-item quotes.
E-kite fiberglass stock	Ø2 mm ×1 m €0.65 incl VAT; Ø3 mm ×1 m €0.80 incl VAT; pultruded UD vinylester. Listing mass differs by variant; analysis uses explicit assumed density/modulus.
Fechtner spring steel	Straightened 1-m stock: Ø1 mm €0.55, Ø2 mm €1.00, Ø3 mm €1.43, VAT included. Catalog straightened is not a measured straightness tolerance.
PB Modélisme aluminium tube	Ø3/2.1 mm ×1 m, wall 0.45 mm; €2.65 single / €2.40 at 10+. Tax treatment not independently confirmed.
MIT structural mechanics, closed-section torsion	Bredt shear flow and closed-section torsion basis; thin-wall equations are a screening approximation for these relatively thick tiny sections.
MIT structural mechanics course notes	Beam bending, open/closed sections and orthotropic-material assumptions.
Jönköping University, brakes and clutches	Cone/annulus normal-force and torque derivation. Does not provide PLA-on-PLA friction values.
MathWorks dog clutch model	Explicit tooth alignment, linkage motion, backlash and contact constraints. Model documentation, not an experimental ban on alternative tooth forms.
KHK gear dimensions and KG Stock Gears technical data	Involute geometry, undercut and contact compatibility. A commercial gear formula does not qualify 0.3-module FDM teeth.
Bambu Studio release v 02.08.02.61	Official Linux binary and preset resources used for the saved-project checks. Tool runtime stays outside the repository.

The E-303 screw model is imported directly, without editing its source or promoting its assumptions into measurements. Its 90° V-thread approximation uses mean diameter 2.3 mm and lead 3 mm; guide opposition and parasitic torque remain assumptions. The selected CAD reuses the actual E-295 retained screw-thread and pin meshes from E-297, preserving their detailed thread profile instead of substituting a nominal cylinder. Only the support/drive end of the new screw changes.

Repository read set: root and experiment/print AGENTS.md; mission/priorities; DES-009 primary document and components; E-295–300; E-303 clutch report/model/scenarios; E-298 owner report including additions; historical CAD generators, dimensions, drawings and print build scripts. No physical results were generated by this investigation.

E-304 — guided 2×2 row/column cone clutch

Status: DRAFT — genuine X1C project saved/reopened/resliced, but retained pin wall paths contain up to 0.207-mm discontinuities at two critical layers. Do not print as a READY job; see the engineering verification record.

Priority: none. **Reviewed:** 2026-10-25.

Source: E-304. DES-009 remains the primary design. No print is scheduled and no purchase is requested.

1. Why this test?

Question: can four properly guided, friction-coupled screw pins engage from stationary arbitrary initial phases, drive from one common input, isolate half-selected neighbors and release at the same follower setting?

The cone has no ideal tooth-phase obstruction. Its printed friction, guide drag and elastic preload cannot be established by CAD. Earlier owner feedback demonstrated wobble, bent selectors and blocked tooth contact, but only one coupled pin. This experiment tests the complete interaction of four cells at the actual 5.08-mm pitch. It does not qualify 80×80 hardware.

Decision: continue only if all four cells repeatedly engage and release without touching their couplers, while an obstructed output does not stop the others. Owner feedback decides whether further work is worthwhile. Stop at binding, visible spring whitening, cracked feet, gear skipping, permanent bending, a stuck cone or unselected pin motion. Do not overcome a jam with pliers or a motor.

2. Prepared print

Primary project: guided-matrix-clutch .3mf. Open it as a project; keep its saved placement and orientation. **Do not auto-orient.** No CAD compilation is required. Review plate map, sections, assembly/exploded view and selection states.

One plate, 33 named objects: one fixed frame; two lower and two upper shared keepers; two row bases and two caps; two columns; four nominal followers plus one thinner and one thicker sample; four nominal input rotors plus one loose and one tight sample; one two-rack shuttle; one hand pinion; four supported output screws in one scaffold object; four terrain pins; one 100-mm beam and two supports. Four screw scaffolds are intentionally separate connected bodies inside one named object. The frame also has two thin sacrificial rods under the stationary follower guides and eight transverse sacrificial rods outside the working raster. Cut their tabs/end contacts and remove all ten rods before assembly; they support print bridges, not the working mechanism. Remove sacrificial screw bases, eight small tabs per screw and side struts after printing. They are not assembled mechanism parts.

Variants are identified by the saved object names and plate map, not tiny unreadable embossed labels. Bag them separately before moving parts: follower-soft t=.35 mm, nominal .40, stiff .45 (depth 3.6); rotor-loose shaftØ1.70, nominal 1.80, tight 1.90. The four nominal copies are the principal 2×2 test. Only change a variant after recording the nominal result.

- **Printer:** Bambu Lab X1 Carbon, 0.4-mm nozzle; Bambu Studio 02.08.02.61. Generic PLA, saved profile inherited from the completed E-295 compact-thread project and official Bambu presets. Exact settings and estimates are in metadata/slicer-review.json.

- **Process:** .12-mm layers and first layer, Arachne, two walls, minimum bead 75%, minimum feature 10%;100% zig-zag infill; nominal line widths 0.45 mm. Outer/inner wall speed 15/20 mm/s; first layer 15; bridge 15 with saved 90° bridge direction and 0% seam gap. Nozzle 220°C, textured PEI bed 55°C. Fan 100% after the first layer, auxiliary fan 0, cooling minimum speed 10 mm/s. Acceleration 1000 general/500 outer and first layer. Supports OFF, raft 0, outer brim 2 mm with .12-mm gap, print by layer. XY contour, hole and elephant-foot compensation 0. Cooling/temperatures and all line widths are retained in the saved project and recorded in the review.

- **Orientation:** row caps and columns roof-down; followers flat back-down so ligaments contact the bed; input rotors cone-tip-down; terrain pins closed-top-down/bore-up; output screws upright cup-up on their supplied scaffold; frame and keeper strips laid on their specified sides; row bases and the 100-mm beam base-down. Threaded cavities have no slicer supports.

- **Estimated time/material:** 7 h 34 min 27 s /21.60 g PLA, from the saved slice; not physical observations.

- **Extra items:** ordinary tape for external keeper/cap tabs, fine marker, ruler; optional scale and known small masses. Rods are **not needed** for the 2×2 test. Optional stock for the beam is twoØ2-mm rods about 100 mm long; record material and diameter. Do not buy rods solely because they appear in CAD.

- **Validation blocker:** the nominal pin bore remains open, but two wall layers have unresolved deposited-path gaps. The file is supplied for review, not as a verified printable pass. See verification. **Validation:** see metadata/guided-matrix-clutch-build.json, geometry-review.json, slicer-review.json and reslice-comparison.json. Printability and mechanism performance remain different evidence levels.

The directory also contains component STLs, parametric CAD in cad/, reproducible build/review tools in scripts/, metadata and SVG illustrations. Engineering STEP/assembly geometry and calculations live under tools/curated-experiment-checks/E-304/ in the repository.

3. Assembly and test

Assemble without forcing fits

Use the actual beam/contact section to identify reinforcement channels and the assembly drawing for the operating frame after removal of sacrificial rods.

1. Remove brims from outer edges, especially the small cone-tip bed contacts. Cut scaffold tabs at the **strut** first; trim the remaining nub carefully. Keep the six-millimetre working thread region intact. Do not sand every thread or cone until it fits: record any required cleanup first.

2. Hand-thread each retained terrain pin onto its screw and check smooth bidirectional motion independently. At neutral the pin's bottom is 6 mm above the thread's bottom. Only use±1.5 mm around that position. Do not run the pin off its guide or to a hard thread stop.

3. Place the two row bases into the frame's bottom longitudinal openings, handles on the same side. Fit each input rotor flange into its follower's open cage before inserting the pair through the row's open side. Place follower lower cross-feet in the row cam tracks. The upper stem stays in the stationary guide. First row opens toward−Y; second toward+Y. Use the exploded drawing to identify the sides.

4. Lower the two cam caps over the feet. The end bridges lie outside the active field. Tape cap to base at the exposed ends; keep tape out of all sliding faces. Confirm the bottom feet are captured but free. Set rows fully OFF; do not compress springs while installing columns.

5. Insert the two column rails around the follower stems, roof above each upper cross-foot. Insert the four pin/screw assemblies into the open output/terrain guides. The output collar belongs **below** the stationary thrust deck. Fit the paired shared keeper strips into their locating slots and retain at external tabs with tape. Confirm all shafts have two supported journal levels. No keeper may lift during operation.

6. Engage both permanent racks with the four input gears and fit the side hand pinion in its frame hole. Initial gear meshing is an assembly step, not clutch engagement. With all selectors OFF, the single knob must rotate all four input rotors while the four output pins remain still. Mark a rotor/cup if motion is hard to see. If a rack catches, a rotor wobbles visibly or an output drags, stop and record the fit before trying variants. The output collar has 0.2-mm axial play; record any height jog on closing. Adverse radial/tilt/axial clearance combinations can rub even though the nominal CAD clears.

The sliders have 2.54-mm travel between corresponding low/high cam dwells. Mark the OFF and ON end positions with a ruler; handles move toward the positive CAD axis for ON. Do not keep moving through successive cam periods. A10° knob notch ideally moves a selected pin 0.25 mm. The delivered fixture is a short-stroke hand demonstration: keep every pin within±1.5 mm of neutral. Mark a rack travel window from−9.5 to+2.8 mm relative to its CAD assembly position (at zero, the rack right end projects about 12.9 mm beyond the frame right edge). Stay inside that window to retain the side-pinion mesh; the knob is not a continuously rotating unlimited-travel drive.

One focused evidence sequence

1. **Logic and isolation:** with input stopped, choose columns, then rows. Check each of the four single cells, each pair in one row, each pair in one column, and all four. Also turn the common input with rows ON/columns OFF and rows OFF/columns ON. Only row-AND-column intersections may move. Record each failed/unknown state; the 16 selector masks are listed in images/selection-states.svg locally as the four cell truth cases and in

metadata/geometry-review.json as full 2×2 masks.

2. **Stationary phase engagement:** set outputs at neutral and the rack about 8.5 mm left of its zero mark. The twelve phase positions then span−8.5 to about+1.87 mm, leaving room for the short forward/back test. With clutches OFF, advance the common knob one 10° notch to change input phase 30°. Hold the knob still, select, then test one notch forward and back. Return selectors OFF. Repeat through 12 input phases spanning a full rotor revolution, initially one cell and then all four. Do not wiggle the knob, push a cone or spin an output to help seating. Record “needed wiggle” as failure of stationary engagement, even if it subsequently drives.

3. **Different phases simultaneously:** use individually selected moves to leave the four outputs at different phase/height offsets (e.g .0, 30, 60, 90°; never exceed stroke limits), then return the shared input to the same mark with clutches OFF. Engage all four from rest and move forward/back one notch. This tests independent contacts rather than four artificially matched pairs.

4. **Load, reversal and release:** first use free-running pins. If a scale is available, place one protruding pin against it with the other pins clear and increase axial resistance gently to approximately 5 g then 20 g indicated (.05/.20 N). Record orientation, actual scale reading and other contact/friction; this is an approximate axial-load test, not a torque measurement. Without a scale report finger resistance as qualitative. At each setting move in both directions, return rows OFF and verify the input turns without moving outputs. Also observe loaded OFF pins for about 10 seconds: no holding brake is included, so gravity-driven descent is possible. Record slow drift; stop for rapid falling or uncontrolled motion. Try a gentle column-only release once and record its force/behavior; normal operation unloads rows before changing columns.

5. **Obstruct one:** restrain one output lightly while all four are selected, turn only far enough to see whether the other three move and the restrained clutch slips. Stop if rack teeth skip, frame flexes, spring whitens or force rises sharply. Return all rows OFF and check release. A blocked output differs from a jammed cam. Do not insert a hard foreign object into the moving rack to simulate a jam.

6. **Repeat and inspect:** at least five cycles per sampled phase if the first pass works (record the actual count). Note settling, wear dust, changed force, permanent follower set and loose keeper/cap tape. Repeat only the failed comparison with one labelled fit/preload variant; changing several variables together obscures the result.

Optional beam check on the same plate: orient the 100-mm beam as in the assembly (deep section vertical), supports 50 then 75 mm apart. Apply a known central load (e.g .100–300 g), measure midpoint deflection relative to unloaded height, and remove load. Repeat with two loose rods in the channels. Record support span, load, measurement resolution and whether any permanent set occurred. A ruler may only bound deflection; do not report unresolvable hundredths of a millimetre. Bonding rods changes the experiment and must be recorded separately. Loose rods do not establish full-composite stiffness.

Single continuation result: all four guided cells repeatedly engage at sampled stationary phases, transmit the chosen light/moderate load in both directions and release at one common follower setting, with half-selected cells still and one obstructed output not stopping its neighbors. That is a physical result to seek, not a pass already obtained. A failed assembly or missing test is also useful evidence; no additional print is automatically authorized.

4. Owner results

Use results.md; blank means unknown. Include the actual printed 3MF if settings/orientation changed. Photos/video are optional.

5. Interpretation after feedback

Physical observations: none yet.

GO / NO-GO: undecided.

Next action: await the owner's explicit decision; no further development, issue, purchase or print campaign is initiated.

E-304 owner results — unperformed

Date:

Actual 3MF / changed settings or orientation:

Printer, nozzle, filament:

Print: [ ] not done [ ] complete [ ] partial [ ] failed

Assembly/cleanup time:

Nominal rotor/follower or named variants:

Missing/damaged parts; scaffold removal/thread scars:

Guide fit, wobble, keeper/cap retention:

Approximate row/column force (measured or felt):

Check	Pass / fail / unknown	Actual observation
All input rotors turn from the one knob with clutches OFF		
Four independent cells		
Both cells in a row / column		
All four simultaneously		
Row-only / column-only half-selection		
12 stationary initial phases, no helping wiggle		
Different output phases simultaneously		
Forward and reverse, free /5 g /20 g or stated load		
Row release / gentle column release		
One restrained output; other three still move		
Loaded OFF retention / drift over 10 seconds		
Wear / spring set after stated number of cycles		

Failed phases (input rotor degrees 0, 30,...330), cell numbers and repetitions:

Measured pin motion vs expected 0.25 mm per knob notch:

Any unselected pin motion:

Anything touched to help engagement (record it):

Optional beam: material/rod diameter, loose/bonded, support span, load, deflection and instrument resolution:

Photo/video link (optional):

Owner decision: [ ] stop [ ] discuss result [ ] explicitly authorize stated next work:

Engineering interpretation: remains blank until actual owner feedback.